

Noise Robustness Mechanisms for Classical Machine Learning in Breast Cancer Diagnosis on WDBC

Sharada S. Weerasinghe
Department of Computer Science & Engineering
University of Moratuwa, Sri Lanka

Abstract—This paper evaluates whether lightweight robustness mechanisms improve classical machine learning for breast cancer diagnosis on the UCI Wisconsin Diagnostic Breast Cancer (WDBC) dataset. Two matched flows are compared under identical cross validation splits and noise schedules: without robustness and with quantile clipping, training jitter, and confidence based sample weighting. The robustness enabled flow improves clean ROC-AUC mean from 0.9935 to 0.9942 (+0.0007), noisy ROC-AUC mean from 0.9919 to 0.9929 (+0.0010), and composite robust score from 0.9904 to 0.9918 (+0.0014). Gains are model dependent, with the largest improvement in Logistic Regression and a small negative gain for Random Forest baseline. The results support lightweight robustness mechanisms as practical reliability improvements for noisy tabular medical classification.

Index Terms—medical machine learning, noise robustness, classical machine learning, breast cancer diagnosis, WDBC, ROC-AUC

I. INTRODUCTION

Reliable diagnostic modeling depends not only on clean data accuracy, but also on stability under data quality degradation. In clinical tabular datasets, two common degradation sources are feature perturbation (measurement variability) and label corruption (annotation or coding errors). Even small noise induced shifts can reduce model trust in medical decision support.

This work investigates a focused question: *do simple noise robustness mechanisms improve classical ML performance compared with an otherwise identical pipeline without those mechanisms?* The study uses WDBC and a strictly matched comparison protocol to isolate mechanism effect from architecture changes.

The scope is intentionally classical (Logistic Regression, SVM-RBF, Random Forest) because these models remain common in tabular healthcare tasks and are practical for resource constrained deployment.

The main contributions are:

- A direct *without vs with* robustness comparison under identical splits, seeds, and noise schedules.
- A compact robustness stack combining quantile clipping, train time jitter, and confidence based sample reweighting.
- Quantitative evidence that robustness mechanisms improve aggregate noise aware performance, with explicit per model gain analysis.

II. RELATED WORK

WDBC is a long standing benchmark for diagnostic classification and was originally introduced for breast cancer diagnosis

from fine needle aspirate features [1]. Reproducibility oriented benchmark suites such as PMLB have reinforced the role of classical tabular models in fair empirical comparison [2], [3]. Core model foundations used here are SVM [4] and Random Forest [5], with implementation through scikit-learn [6].

Unlike studies that introduce new model families, this paper evaluates whether practical robustness controls improve reliability within a fixed classical ML setup.

III. DATASET AND PROBLEM SETUP

A. Dataset

We use the UCI WDBC dataset [1], with 569 samples and 30 numeric predictors. Features are derived from ten nucleus shape/texture descriptors using mean, standard error, and worst value summaries. The target is binary diagnosis (benign or malignant). The ID field is removed, and diagnosis is mapped to {0,1}.

B. Model Presets

Six presets are evaluated. Logistic Regression uses a baseline preset (StandardScaler, $C = 1.0$) and a robust preset (RobustScaler, $C = 0.3$). SVM-RBF uses a baseline preset (StandardScaler, $C = 1.0$) and a robust preset (RobustScaler, $C = 0.8$). Random Forest uses a baseline preset ($n_{trees} = 500$, $min_samples_leaf = 1$) and a robust preset ($n_{trees} = 800$, $min_samples_leaf = 4$, class balanced subsampling).

C. Noise Protocol

We evaluate two controlled noise families. Feature noise uses Gaussian perturbation scale $\alpha \in \{0.00, 0.05, 0.10, 0.20, 0.30\}$. Label noise uses random label flip rate $p \in \{0.00, 0.02, 0.05, 0.10, 0.15\}$.

IV. METHODOLOGY

A. Comparison Flows

The paper compares two flows:

- **Without robustness:** quantile clipping off, training jitter off, confidence weighting off.
- **With robustness:** quantile clipping on, training jitter $\alpha = 0.02$, confidence weighting on.

All other settings (data split, seed, model presets, noise levels, metrics) are kept identical.

TABLE I
OVERALL GAIN FROM ENABLING ROBUSTNESS MECHANISMS.

Metric	Without	With	Gain
Clean ROC-AUC mean	0.9935	0.9942	+0.0007
Noise ROC-AUC mean	0.9919	0.9929	+0.0010
Robust score mean	0.9904	0.9918	+0.0014

TABLE II
CLEAN DATA SUMMARY (MEAN ACROSS FOLDS).

Model	Accuracy	F1 (Malignant)	ROC-AUC
LogReg (Baseline)	0.9772	0.9685	0.9960
LogReg (Robust)	0.9754	0.9670	0.9965
SVM-RBF (Baseline)	0.9737	0.9641	0.9952
SVM-RBF (Robust)	0.9701	0.9592	0.9954
RF (Baseline)	0.9561	0.9400	0.9908
RF (Robust)	0.9543	0.9395	0.9913

B. Robustness Mechanisms

Quantile clipping limits each feature to the train fold 1st–99th percentile range, reducing outlier leverage.

Training jitter adds low scale Gaussian perturbation to training features, promoting smoother decision boundaries.

Confidence weighting detects low confidence samples with an internal train fold detector and down weights suspected noisy labels during model fitting.

C. Validation and Fairness Controls

Main comparison runs use repeated stratified 5 fold cross validation with fixed seed 42 ($n_splits = 5$, $n_repeats = 1$). To prevent leakage:

- clipping thresholds are computed from train folds only,
- confidence weighting is computed from train data only,
- noise protocols and split definitions are matched across the two comparison flows.

D. Metrics and Robust Score

Per fold metrics are Accuracy, Precision (malignant), Recall (malignant), Specificity (benign), F1 (malignant), and ROC-AUC. The noise aware score is

$$J = \mu_{\text{noise}} - \lambda \sigma_{\text{noise}}, \quad \lambda = 0.5, \quad (1)$$

where μ_{noise} and σ_{noise} are mean and standard deviation of ROC-AUC across noisy scenarios.

V. EXPERIMENTAL RESULTS

A. Overall Comparison

Table I reports the primary comparison. Robustness mechanisms improve all aggregate indicators.

B. Clean Data Baseline Context

Table II provides clean data context for the six presets. All presets perform strongly (ROC-AUC > 0.99), which indicates a high baseline ceiling and makes robustness gains inherently incremental.

TABLE III
PER MODEL GAIN FROM ROBUSTNESS MECHANISMS.

Model	Robust Score Gain	Drop Reduction
LogReg (Baseline)	+0.0038	+0.0017
LogReg (Robust)	+0.0030	+0.0010
SVM-RBF (Robust)	+0.0013	+0.0009
SVM-RBF (Baseline)	+0.0012	+0.0004
RF (Robust)	+0.0002	-0.0003
RF (Baseline)	-0.0009	-0.0016

TABLE IV
ROBUSTNESS RANKING WITH ROBUSTNESS ENABLED.

Model	Clean ROC-AUC	Noisy ROC-AUC Mean	Robust Score
SVM-RBF (Robust)	0.9954	0.9951	0.9947
SVM-RBF (Baseline)	0.9952	0.9949	0.9945
LogReg (Robust)	0.9965	0.9948	0.9939
LogReg (Baseline)	0.9960	0.9933	0.9915
RF (Robust)	0.9913	0.9903	0.9893
RF (Baseline)	0.9908	0.9891	0.9872

C. Per Model Gain and Ranking

Table III and Fig. 1 show model specific gains. Logistic Regression benefits the most, SVM improves moderately, and RF baseline shows a small negative gain.

D. Behavior Under Feature Noise

Fig. 2 shows mean ROC-AUC across model presets versus feature noise level. The robustness enabled flow stays above the non robust flow at almost all levels and the separation becomes clearer in higher noise settings.

E. Behavior Under Label Noise

Fig. 3 shows a larger separation under label noise than under feature noise, indicating that label corruption is the stronger failure mode for this setup. Robustness mechanisms preserve a higher ROC-AUC trajectory as p increases.

F. Robustness Objective Ranking

Table IV presents ranking under the *with robustness* flow. SVM variants lead in robustness score due to strong mean noisy ROC-AUC with low variance.

VI. DISCUSSION

The primary objective is met: enabling robustness mechanisms improves aggregate clean performance, aggregate noisy performance, and composite robust score (Table I).

Model family behavior is non uniform (Table III). Logistic Regression benefits strongly, SVM improves moderately and consistently, and Random Forest baseline can lose performance under the shared configuration. This indicates that a global robustness recipe is useful, but still requires family specific calibration.

The noise trajectories in Fig. 2 and Fig. 3 indicate that label noise is the stronger stressor in this task. When annotation uncertainty is expected, confidence weighted training and label quality controls should be prioritized.

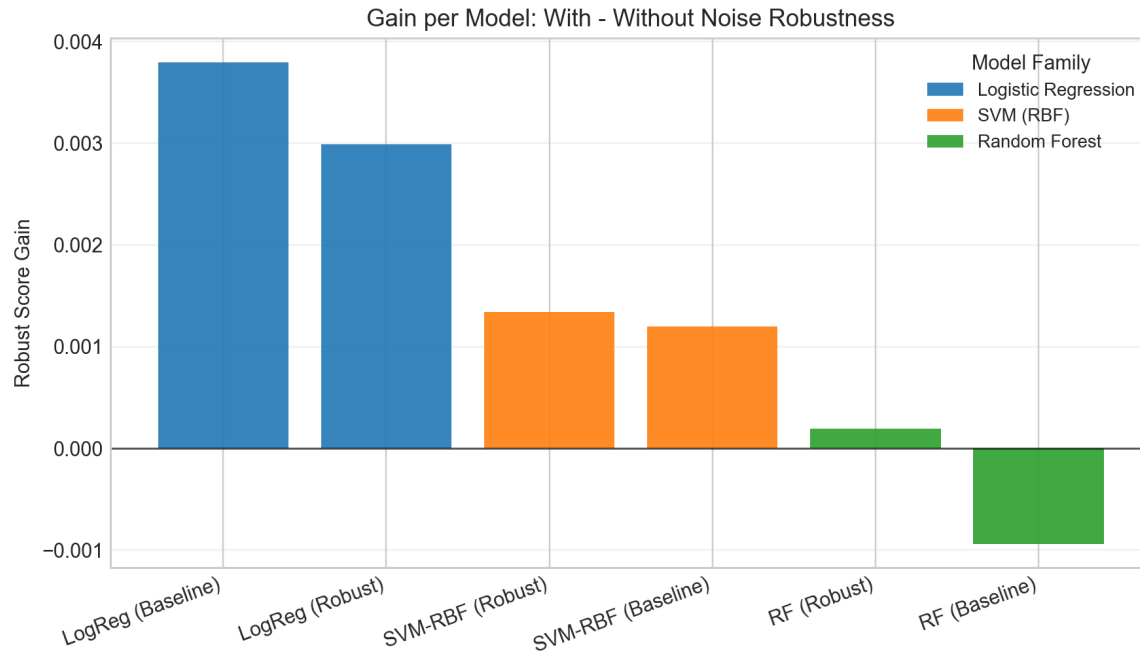


Fig. 1. Robust score gain per model (with minus without). Five of six presets improve; RF baseline is the only negative case.

A. Practical Implications

The magnitude of aggregate gain is modest but operationally meaningful in high reliability domains. For this study, robustness mechanisms produce consistent improvements without introducing deep architectures or expensive retraining schedules. The approach is therefore suitable for baseline clinical decision support pipelines where interpretability and computational cost are key constraints.

VII. THREATS TO VALIDITY AND LIMITATIONS

This work uses one dataset and fixed presets. While the matched comparison design is strong for internal validity, external generalization is limited.

Key limitations are:

- no formal significance tests on fold level differences,
- no mechanism level ablation to isolate clipping/jitter/weighting contributions,
- single dataset scope without cross dataset transfer checks.

These are planned as future extensions and are separated from the completed core comparison in this paper.

VIII. CONCLUSION

This paper presented a classical ML robustness study for breast cancer diagnosis on WDBC. Under a matched comparison protocol, enabling quantile clipping, train jitter, and confidence weighting improved clean ROC-AUC mean by +0.0007, noisy ROC-AUC mean by +0.0010, and robust score mean by +0.0014. Gains are model dependent, with strongest improvements in Logistic Regression and limited benefit in Random Forest baseline.

The results support using lightweight robustness mechanisms as a reliability layer in tabular medical classifiers, while maintaining model specific tuning as a standard practice.

REPRODUCIBILITY NOTE

All numbers and figures in this manuscript are produced from project outputs in the results and plots directories generated by the notebook workflow.

REFERENCES

- [1] W. H. Wolberg, W. N. Street, and O. L. Mangasarian, "Breast cancer wisconsin (diagnostic) data set," UCI Machine Learning Repository, 1995, accessed: 2026-04-10.
- [2] R. S. Olson, W. L. Cava, Z. Mustahsan, A. Varik, and J. H. Moore, "Pmlb: a large benchmark suite for machine learning evaluation and comparison," *BioData Mining*, vol. 10, no. 1, p. 36, 2017.
- [3] J. D. Romano *et al.*, "Pmlb v1.0: an open source dataset collection for benchmarking machine learning methods," 2021.
- [4] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [5] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [6] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

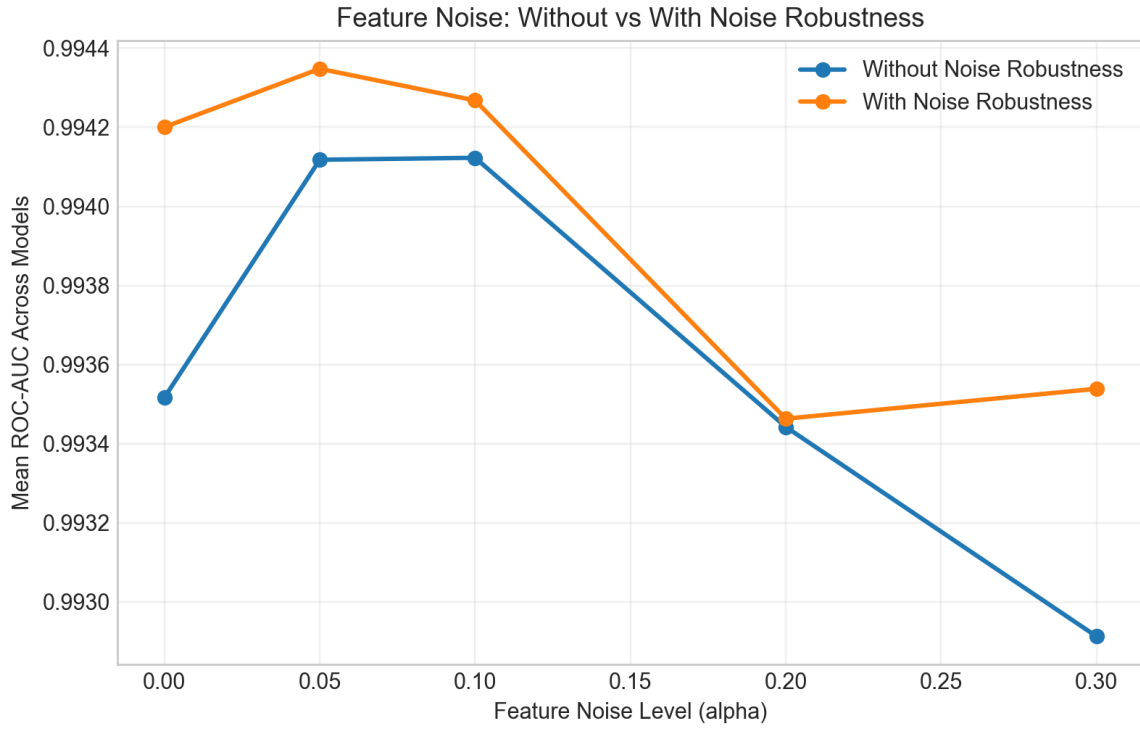


Fig. 2. Mean ROC-AUC under feature noise. The robustness enabled flow remains above the non robust flow as α increases.

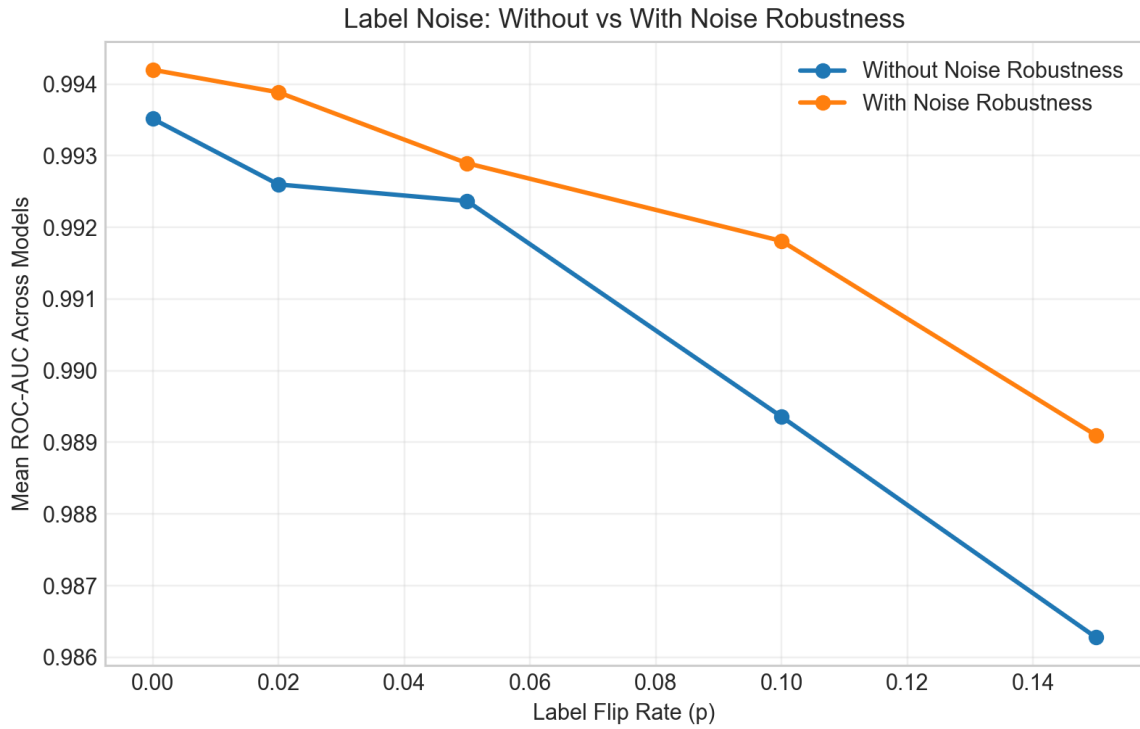


Fig. 3. Mean ROC-AUC under label noise. Separation between flows grows with label flip rate p .